



02451 Introduction to Machine Learning

Week 7: Performance evaluation and AUC

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17 March 2025

DTU Compute, Technical University of Denmark

Today

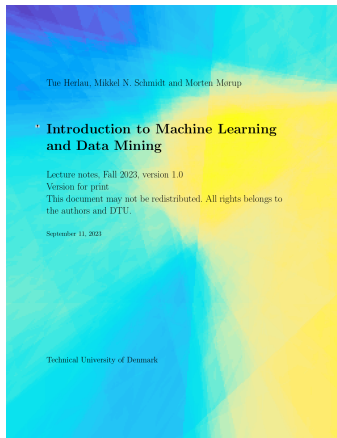
Feedback Groups of the day:

Sebastian Thor Dollerup Bech, Victor Birkefeldt Carstens, Mikkel Baun Christensen, Mark Djatsuk, Muhammad Muneeb Farooq, Liva Frandsen-Melau, Maria Frazão, Anton Valerio Greve, Thomas Hedegaard, Albert Bonde Hjortkjær, Freya Drøschler Holm, Juhani Petteri Jokio, Timon Kun, Dejan Joel Künnemeyer, Ze Heng Lai, Sara Kristine Majholm, Carl Johann Creutzberg Mikkelsen, Rasmus Holmgaard Nielsen, Jens Palmer Agnero Olsen, Theodoros Papanikolas, Anders Helmuth Rame, Rasmus Emil Fabæch Brøndby Reimer, Asger Røgind, Markus Emil Aakjær Saxe, Bertram Lillelund Sillesen, Julius Terp, David Tetteh, Vishnu Venkatesh, Roald Thomas Waltenburg.

Reading/homework material:

Chapter 11, 16

S24-Q26, F24-Q24, P16.1, P16.2



Lecture Schedule

- 1** Introduction, data, and visualization
3 February: C1, C2, C7
- 2** Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3** Computational linear algebra and PCA
17 February: C3
- 4** Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5** Bayesian inference and linear models
3 March: C5, C6, C8
- 6** Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7** Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8** Artificial Neural Networks and Optimization
24 March: C15
- 9** Bias/Variance and ensemble methods
7 April: C14, 17
- 10** K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11** Mixture models and density estimation
21 April: C19, C20
- 12** Representation learning
28 April: Material provided separately
- 13** Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Practicalities and announcements

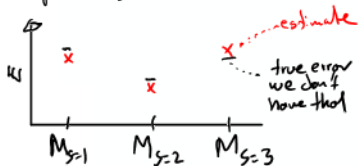
- Results of the first batch of assignments
 - **Submit BOTH html and ipynb!**
- Only handwritten notes (not printed) are allowed during the exam (i.e., **no** tablet/computer-printed notes).
- Deadline W4-b assignments 19/3 17:00
- Midterm test next week (after deadline for W4-b)
- 6.2 : argmin vs argmax

Recap Lecture 6

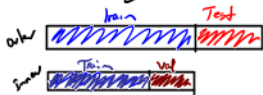
- Given a model M , estimate the generalization error r



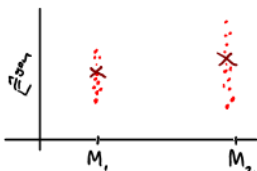
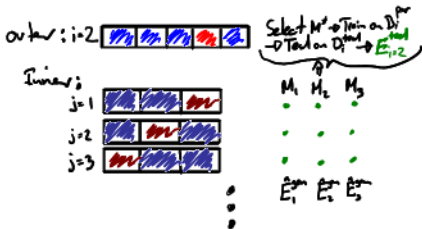
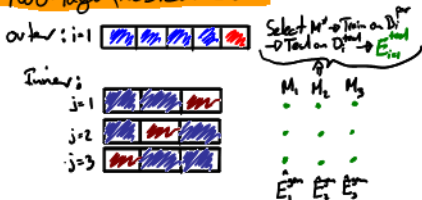
- Given many models (e.g. different complexities) Select the best one



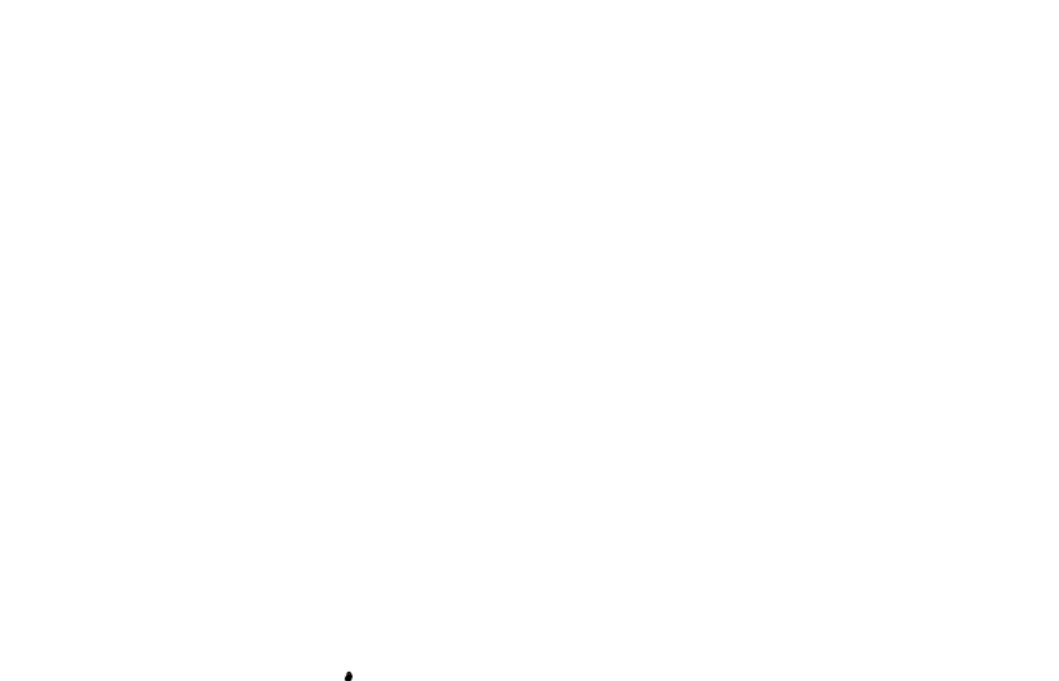
- ... but also report a "good" estimate of the generalization error



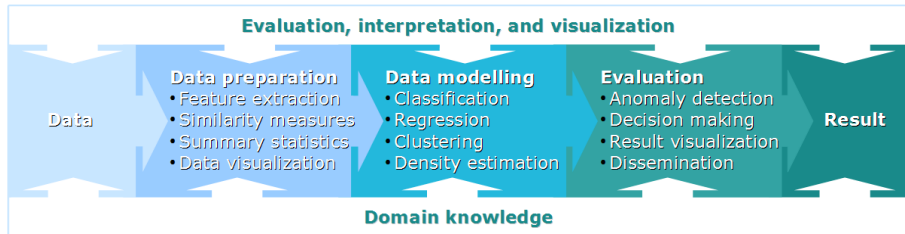
Two-layer/nested CV



- a stochastic estimate
- How do we
 - 1) Do estimation of E^{train}
 - 2) Compare M_1 and M_2
 ... while taking the uncertainty/variability into account?



Learning Objectives



Learning Objectives

- Understand the two different evaluation setups
- Apply appropriate statistical tests to evaluate and compare models
- Be able to address issues of class-imbalance and resampling
- Understand the definition of Precision, Recall, ROC, and AUC

Statistical testing

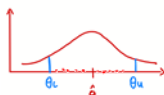
Statistical testing

- A social media company wish to know if a new ad-placement method increases the click-through rate
- How many customers are likely to click adds next month?
- How well can a logistic regression model learn to distinguish between diseased/non-diseased X-rays?
- Should I recommend my logistic regression model over a competing method?

All involve induction beyond the dataset

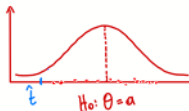
Statistical testing

Confidence interval



- collect a (random) dataset
- assume distribution
- estimate parameters from data

Hypothesis testing



- collect a (random) dataset
- assume distribution
- check if statistic from data is plausible.

Tests **can** provide:

- An objective way to choose between methods
- A quantification of model performance which takes uncertainty into account

Tests **do not** provide

- Certain conclusions (Model A is better than B)
- A black-box recipe

Use statistical tests to aid your interpretation of your results **not** as an argument in itself

Outline

- What is our overall **objective**? What conclusions do we want?
- What tools do we have available?
 - Estimation (values and confidence intervals)
 - Hypothesis testing (eg. model A better than model B)
- What specific test should I use?
 - Classification (e.g. McNemar)
 - Regression (e.g. Paired t-test)

The **objective** and **evaluation criteria**

- Models are compared based on how well they **generalize to future data**
- Suppose we have data $\mathcal{D} = (\mathbf{X}, \mathbf{y})$ and two models $\mathcal{M}_A, \mathcal{M}_B$
- Training on $\mathcal{D}$, we obtain prediction rules

$$f_{\mathcal{D},A} : \mathbf{x} \rightarrow y, \quad \text{and} \quad f_{\mathcal{D},B} : \mathbf{x} \rightarrow y.$$

- Compared by the **difference in generalization error**:

$$z_{\mathcal{D}} = E_{\mathcal{D},A}^{\text{gen}} - E_{\mathcal{D},B}^{\text{gen}}$$
$$E_{\mathcal{D},A}^{\text{gen}} = \int p(\mathbf{x}, y) L(f_{\mathcal{D},A}(\mathbf{x}), y) d\mathbf{x} dy, \quad E_{\mathcal{D},B}^{\text{gen}} = \int p(\mathbf{x}, y) L(f_{\mathcal{D},B}(\mathbf{x}), y) d\mathbf{x} dy.$$

- If $z_{\mathcal{D}} < 0$, it means **that $\mathcal{M}_A$ is better than $\mathcal{M}_B$when trained on $\mathcal{D}$**

Setup I Statistical tests of performance considering the **specific** training set $\mathcal{D}$

An alternative objective

$$z_{\mathcal{D}} = E_{\mathcal{D},A}^{\text{gen}} - E_{\mathcal{D},B}^{\text{gen}}$$

- If you prove $z_{\mathcal{D}} < 0$, you can't know if this is true for an independent dataset $\mathcal{D}'$
- Therefore, the conclusion is not independently reproducible
- To overcome this, test if $\mathcal{M}_A$ is better than $\mathcal{M}_B$ when averaging over dataset

$$z = \mathbb{E}_{\mathcal{D}}[z_{\mathcal{D}}] < 0$$

$$E^{\text{gen}} = \int \left[\int L(f_{\mathcal{D}}(\mathbf{x}), y) p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y} \right] p(\mathcal{D}) d\mathcal{D}$$

- If $z < 0$, it means $\mathcal{M}_A$ is better than $\mathcal{M}_B$... using a typical training set

Setup II *Statistical tests of performance considering a dataset of size N*

Choices, choices

Setup I Statistical tests of performance considering the **specific** training set $\mathcal{D}$?

Setup II *Statistical tests of performance considering a **dataset** of size N*

Which to choose fundamentally depends on what you want to conclude

- Setup II is a more general (impressive) conclusion
- Setup II is probably what we want in science
- Setup II requires (a lot of) cross-validation
- If you have a single train/test split, use setup I

We will consider **setup I** here

Statistical tasks and tools

Let z be a quantity of interest (for instance $z = E_{\mathcal{A}}^{\text{gen}} - E_{\mathcal{B}}^{\text{gen}}$)

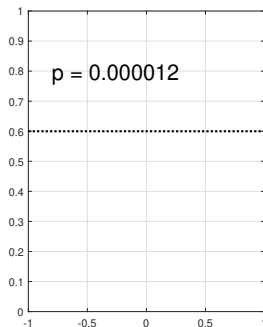
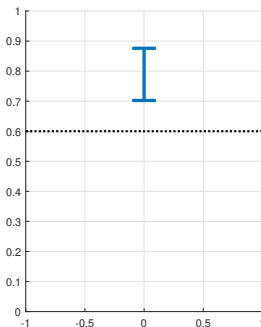
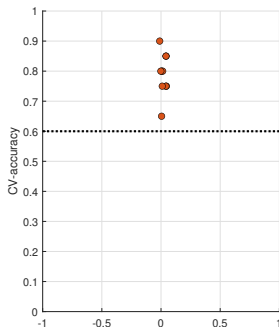
Hypothesis testing Determine **whether there is an effect** by choosing between
 $H_0 : z = 0$ vs. $H_1 : z \neq 0$

Estimation Determine a likely value $z \approx \hat{z}$ and **an interval** $[z_L, z_U]$ that likely contains z

- Evidence against H_0 is measured by a **p -value** (low p is evidence for an effect $z \neq 0$)
- Estimation of $[z_L, z_U]$ done using an **α -confidence interval** (lower α means a more conservative, wider, interval)

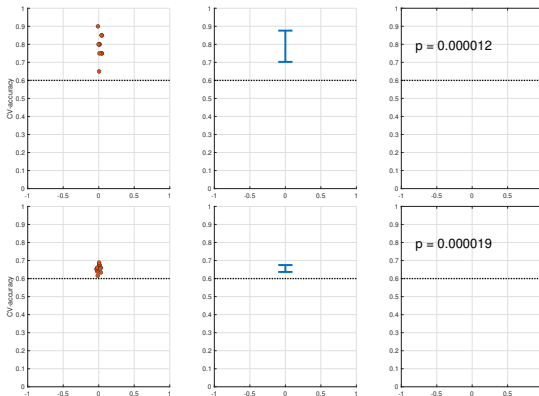
Choosing the right tool

- Consider binary classification using $N = 200$ samples
- We estimate test error using $K = 10$ -fold CV (10 test-error estimates)
- Question: Is accuracy $E_A^{\text{gen}} \approx \frac{1}{K} \sum_{k=1}^K E_i^{\text{test}}$ greater than baseline θ_0 ?
- (Baseline classify everything as maximum class, accuracy 60%)



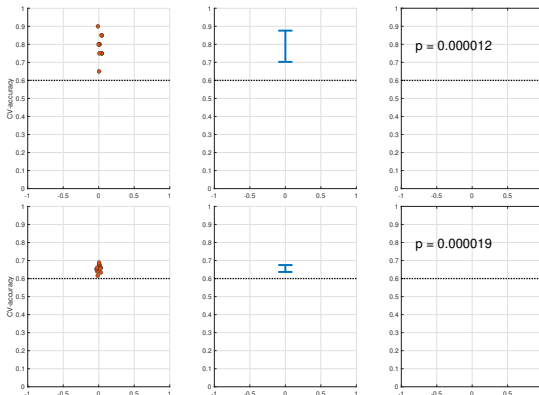
Which tool to use

- Top: $N = 200$ sample example
- Bottom: Harder problem using $N = 2000$ samples



Which tool to use

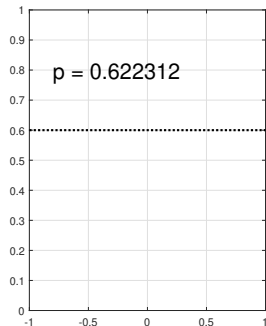
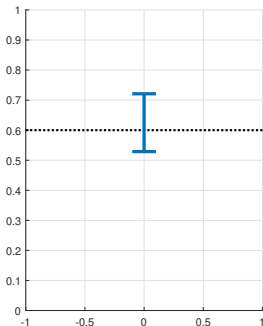
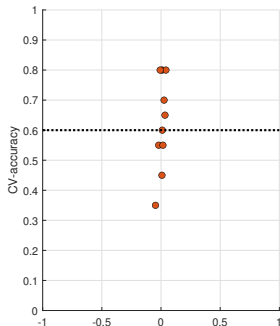
- Top: $N = 200$ sample example
- Bottom: Harder problem using $N = 2000$ samples



- p -value primarily measure of sample size (not **effect size!**)
- Which do **you** think are more informative?

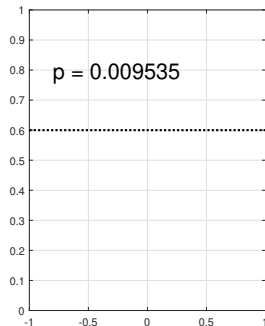
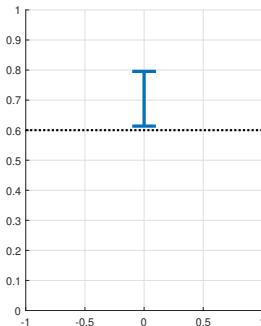
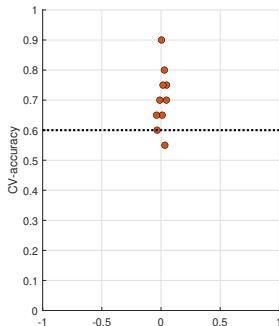
Variability

- New problem using $N = 200$ samples. Is there an effect?

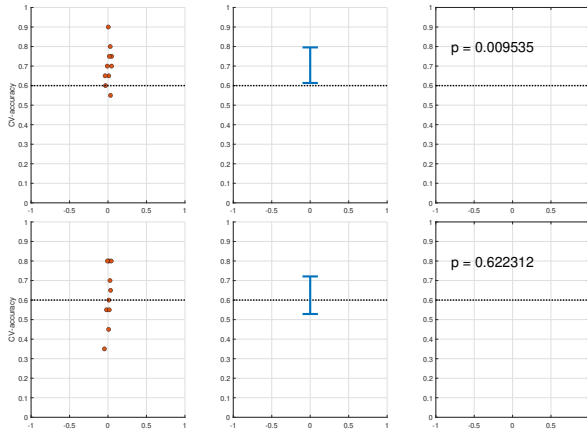


Variability

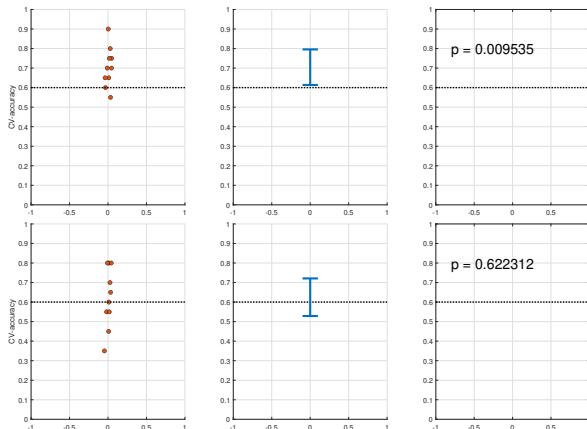
- Another problem using $N = 200$ samples. Is there an effect?



The nasty bit



The nasty bit



- Only difference is random variability in dataset
- Low p -value does **not necessarily** mean reproducible
 - Training many models will lead to false positives
 - Statistics **will not** fix unclear results: probably just lead to false positives

Connecting objective to numbers

- We want to draw conclusions about the difference in performance:

$$z_{\mathcal{D}} = E_{\mathcal{D},A}^{\text{gen}} - E_{\mathcal{D},B}^{\text{gen}}$$

$$E_{\mathcal{D},A}^{\text{gen}} = \int p(\mathbf{x}, y) L(f_{\mathcal{D},A}(\mathbf{x}), y) d\mathbf{x} dy, \quad E_{\mathcal{D},B}^{\text{gen}} = \int p(\mathbf{x}, y) L(f_{\mathcal{D},B}(\mathbf{x}), y) d\mathbf{x} dy.$$

- This can be estimated as

$$\begin{aligned} \hat{z}_{\mathcal{D}} &= \frac{1}{N^{\text{test}}} \sum_{i=1}^{N^{\text{test}}} [L(f_{\mathcal{D},A}(\mathbf{x}_i), y_i) - L(f_{\mathcal{D},B}(\mathbf{x}_i), y_i)] \\ &= \frac{1}{N^{\text{test}}} \sum_{i=1}^{N^{\text{test}}} z_i, \quad \text{where:} \quad z_i = L(f_{\mathcal{D},A}(\mathbf{x}_i), y_i) - L(f_{\mathcal{D},B}(\mathbf{x}_i), y_i). \end{aligned}$$

Abstracting to a statistical question

Consider data as the n numbers

$$D = (z_1, \dots, z_n).$$

General form of the problem: Draw conclusions about

$$\theta = E_{A,D}^{\text{gen}} - E_{B,D}^{\text{gen}}$$

Based on the estimate:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n z_i.$$

Example:

Test scores before and
after attending a
TA session

A: [10 12 2 20]

B: [5 1 10 5]

D = [5 11 -9 15]



(1 1 0 1 1 0 0 1 1)
}

Statistical tools: Parameter

- Assume z_i is a realization of a random variable Z_i
- It has density

$$p(Z_i = z_i | \theta) = p_\theta(z_i)$$

$$e.g. = \mathcal{N}(z_i | \mu = \theta, \sigma^2)$$

$$e.g. = \theta^{z_i} (1 - \theta)^{(1 - z_i)}$$

$$e.g. = \mathcal{N}_n(z_i | \mu = \theta, \sigma^2)$$

$$p_\theta(D) = \prod_{i=1}^n p_\theta(z_i).$$

$$e.g. = \theta^m (1 - \theta)^{n - m}$$

- Returning to our goals:
 - estimating plausible ranges of θ**
 - hypothesis testing such as whether θ takes a particular value**
- Let's look at the statistical tools to accomplish this

Examples:

- Height of students in cm
- Classifier A correct (0/1)
- Difference in test error for regression model A and B

Statistical tools: Statistic and estimator

Statistic A statistic is a function of the data D and will be denoted t . For instance, the mean and variance are both statistics:

$$t_0(D) = \frac{1}{n} \sum_{i=1}^n Z_i, \text{ or } t_1(D) = \frac{1}{n} \sum_{i=1}^n (Z_i - t_0(D))^2.$$

Estimator An estimator is a statistic t of D such that $t(D)$ is close to θ . In the examples we will consider the mean

$$t_0(D) = \frac{1}{n} \sum_{i=1}^n Z_i$$

Statistical tools: Confidence interval

- A **confidence interval** (CI) is an interval $[\theta_L, \theta_U]$ which likely contains θ
- The CI is a function of the data D . θ_L and θ_U are two statistics and for a concrete dataset the interval is computed to be

$$[\theta_L(D), \theta_U(D)].$$

- With probability $1 - \alpha$, the true value θ should fall within the confidence interval $[\theta_L(D), \theta_U(D)]$ as we randomize over different datasets

$$P_{\theta}(\theta \in [\theta_L, \theta_U]) = 1 - \alpha.$$



$$z_i = E_n - E_B$$

Statistical tools: Null hypothesis testing and p -value

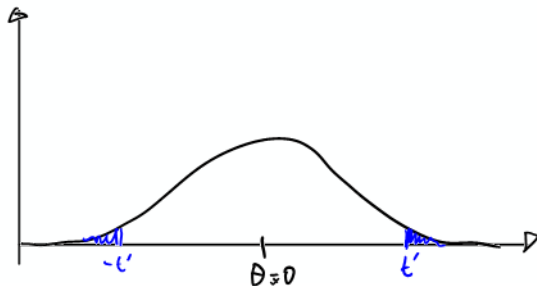
- Determining whether a **null hypothesis** H_0 about the parameters is true or false

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta \neq 0$$

- Intuitively, if H_0 is true, the data should behave in a certain way

- We test if the data is implausible assuming H_0**

$$t(D) = \frac{1}{n} \sum_i z_i, \quad t = \frac{1}{n} \sum_i z_i \quad p(t(D) = t | H_0)$$



Low p -value

- we reject the H_0
- does not mean H_1 is true
- rejection due to: evidence wrong assumptions etc.

High p -value:

- we cannot reject H_0
- does not mean H_0 is true
- fail to reject...

Statistical tools: Null hypothesis testing and p -value

- Determining whether a **null hypothesis** H_0 about the parameters is true or false

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta \neq 0$$

- Intuitively, if H_0 is true, the data should behave in a certain way
 - **We test if the data is implausible assuming H_0**
- Specifically, let t be a statistic, for our purpose

$$t(D) = \frac{1}{n} \sum_{i=1}^n Z_i$$

On our dataset it has a particular value $t_0 = \frac{1}{n} \sum_{i=1}^n z_i$

- We can compute the density $t(D)$ takes a particular value given H_0 is true:

$$p(t(D) = t | H_0) = p_{\theta=\theta_0}(t(D) = t)$$

- p -value is the chance $t(D)$ is at least as extreme as what we actually observed:

$$p\text{-value} : p = P(t(D) > |t_0| \mid H_0) = P_{\theta=\theta_0}(t(D) \geq |t_0|).$$

Setup I: Fixed training set

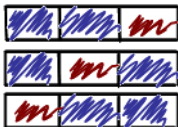
Suppose we carry out cross-validation to obtain:

$$(\mathcal{D}_1^{\text{train}}, \mathcal{D}_1^{\text{test}}), \dots, (\mathcal{D}_K^{\text{train}}, \mathcal{D}_K^{\text{test}}).$$

We collect these into (paired) vectors of predictions and true values:

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_K \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1^{\text{test}} \\ y_2^{\text{test}} \\ \vdots \\ y_K^{\text{test}} \end{bmatrix}.$$

Eg 3-fold-classification



$$\mathbf{y} = [0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \ 1]$$

$$\hat{\mathbf{y}} = [1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 1]$$

$$\mathbf{D} = (0, 1, 1, 1, 1, 0, 0, 1, 1)$$

Evaluation of a **single classifier**

- Define:

$$c_i = \begin{cases} 1 & \text{if } \hat{y}_i = y_i \\ 0 & \text{if otherwise.} \end{cases}$$

E.g.

$$D = (0, 1, 1, 1, 1, 0, 0, 1, 1, 1)$$

- Number of accurate guesses:

$$m = \sum_{i=1}^n c_i.$$

Recall

$$p(D|\theta) = \prod_{i=1}^n \theta^{c_i} (1-\theta)^{(1-c_i)} \\ = \theta^m (1-\theta)^{n-m}$$

- Let the chance the classifier is correct be θ . Then, from **Lecture 5**, we know (assume $a=b=\frac{1}{2}$)

$$p(\theta|m, n) = \text{Beta}(\theta|a, b), \quad a = m + \frac{1}{2}, \text{ and } b = n - m + \frac{1}{2}.$$

Evaluating a **single classifier** (Jeffreys interval)

- If m is the number of accurate guesses, then

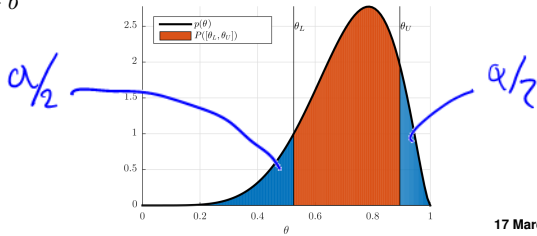
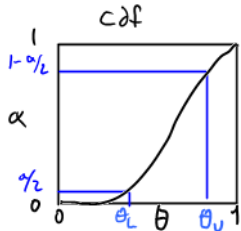
$$p(\theta|m, n) = \text{Beta}(\theta|a, b), \quad a = m + \frac{1}{2}, \text{ and } b = n - m + \frac{1}{2}.$$

- The $1 - \alpha$ confidence interval is given as $[\theta_L, \theta_U]$:

$$\theta_L = \text{cdf}_B^{-1}\left(\frac{\alpha}{2}|a, b\right) \text{ if } m > 0 \text{ otherwise } \theta_L = 0$$

$$\theta_U = \text{cdf}_B^{-1}\left(1 - \frac{\alpha}{2}|a, b\right) \text{ if } m < n \text{ otherwise } \theta_U = 1$$

$$\hat{\theta} = \mathbb{E}[\theta] = \frac{a}{a+b}$$



Comparing two classifiers

- Assume we have predictions from both classifiers:

$$\hat{\mathbf{y}}^A = \hat{y}_1^A, \dots, \hat{y}_n^A, \quad \hat{\mathbf{y}}^B = \hat{y}_1^B, \dots, \hat{y}_n^B.$$

- As before, we want to know if the classifiers are correct or not:

$$c_i^A = \begin{cases} 1 & \text{if } \hat{y}_i^A = y_i \\ 0 & \text{if otherwise.} \end{cases} \quad \text{and} \quad c_i^B = \begin{cases} 1 & \text{if } \hat{y}_i^B = y_i \\ 0 & \text{if otherwise.} \end{cases}$$

- The relevant information is the contingency table:

$$n_{11} = \sum_{i=1}^n c_i^A c_i^B = \{\text{Both classifiers are correct}\}$$

$$n_{12} = \sum_{i=1}^n c_i^A (1 - c_i^B) = \{A \text{ is correct, } B \text{ is wrong}\}$$

$$n_{21} = \sum_{i=1}^n (1 - c_i^A) c_i^B = \{A \text{ is wrong, } B \text{ is correct}\}$$

$$n_{22} = \sum_{i=1}^n (1 - c_i^A) (1 - c_i^B) = \{\text{Both classifiers are wrong}\}$$

Comparing **two classifiers**: McNemar's test

- We want to compare the accuracy difference: $\theta = \theta_A - \theta_B$
- It is possible to show (approximately)

$$p(\theta|\mathbf{n}) = \frac{1}{2} \text{Beta} \left(\frac{\theta + 1}{2} \mid a = f, b = g \right),$$

$$f = \frac{E_\theta + 1}{2} (Q - 1) \quad g = \frac{1 - E_\theta}{2} (Q - 1)$$

$$E_\theta = \frac{n_{12} - n_{21}}{n}, \quad Q = \frac{n^2(n+1)(E_\theta + 1)(1 - E_\theta)}{n(n_{12} + n_{21}) - (n_{12} - n_{21})^2}.$$

$$\theta_L = 2\text{cdf}_B^{-1} \left(\frac{\alpha}{2} \mid a = f, b = g \right) - 1, \quad \theta_U = 2\text{cdf}_B^{-1} \left(1 - \frac{\alpha}{2} \mid a = f, b = g \right) - 1$$

- For a p -value, note that A is better than B if $n_{12} > n_{21}$
- A p -value can be obtained as:

$$p = 2\text{cdf}_{\text{binom}} \left(m = \min\{n_{12}, n_{21}\} \mid \theta = \frac{1}{2}, N = n_{12} + n_{21} \right)$$

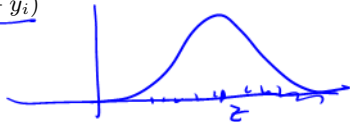


Confidence interval for a **single regression model**

- Use cross-validation to obtain predictions $\hat{y}_i$ and true values y_i . Select loss

$$z_i = |\hat{y}_i - y_i| \quad \text{or} \quad z_i = (\hat{y}_i - y_i)^2$$

- Estimated error is: $\hat{z} = \frac{1}{n} \sum_{i=1}^n z_i$.
- Assume** each error is normally distributed

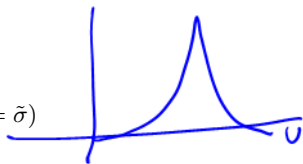


$$p(D|u, \sigma^2) = \prod_{i=1}^n \mathcal{N}(z_i|u, \sigma^2)$$

- Now u follows a generalized Student's t -distribution:

$$p(u|D) = p_{\mathcal{T}}(u|\nu = n - 1, \mu = \hat{z}, \sigma = \tilde{\sigma})$$

with parameters $\hat{z} = \frac{1}{n} \sum_{i=1}^n z_i$ and $\tilde{\sigma} = \sqrt{\sum_{i=1}^n \frac{(z_i - \hat{z})^2}{n(n-1)}}$.



- The Student's t -distribution has density

$$p_{\mathcal{T}}(x|\nu, \mu, \sigma) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\pi\nu\sigma^2}} \left(1 + \frac{1}{\nu} \left[\frac{x - \mu}{\sigma}\right]^2\right)^{-\frac{\nu+1}{2}}.$$

Confidence interval for a **single regression model**

- Step back: Assuming $z_i = L(y_i, \hat{y}_i)$ and

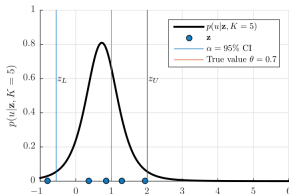
$$z_i \sim \mathcal{N}(z_i | \mu = u, \sigma^2)$$

- In this case u is the average error rate. Since we have shown:

$$p(u|D) = p_{\mathcal{T}}(u | \nu = n - 1, \mu = \hat{z}, \sigma = \tilde{\sigma})$$

- An approximate $1 - \alpha$ confidence interval is:

$$z_L = \text{cdf}_{\mathcal{T}}^{-1} \left(\frac{\alpha}{2} \mid \nu, \hat{z}, \tilde{\sigma} \right), \quad z_U = \text{cdf}_{\mathcal{T}}^{-1} \left(1 - \frac{\alpha}{2} \mid \nu, \hat{z}, \tilde{\sigma} \right).$$



Comparing **two regression models**

- Use cross-validation to obtain (paired) predictions along with true values y_i

$$\hat{y}_1^A, \dots, \hat{y}_n^A, \quad \text{and} \quad \hat{y}_1^B, \dots, \hat{y}_n^B.$$

- Select a loss-function to compute the per-observation losses as in

$$z_1^A, \dots, z_n^A, \quad \text{and} \quad z_1^B, \dots, z_n^B.$$

- Note that

$$\begin{aligned} z &= E_{A,\mathcal{D}}^{\text{gen}} - E_{B,\mathcal{D}}^{\text{gen}} \approx \hat{z} = \left(\frac{1}{n} \sum_{i=1}^n z_i^A \right) - \left(\frac{1}{n} \sum_{i=1}^n z_i^B \right) \\ &= \frac{1}{n} \sum_{i=1}^n z_i, \quad \text{where } z_i = z_i^A - z_i^B \end{aligned}$$

- Assume $z_i \sim \mathcal{N}(z_i | \mu = u, \sigma^2)$
- Compute a $1 - \alpha$ CI using methods on previous slide

Comparing **two regression** models: p -values

$$z = E_A^{\text{gen}} - E_B^{\text{gen}} \approx \hat{z} = \frac{1}{n} \sum_{i=1}^n z_i, \quad \text{where } z_i = z_i^A - z_i^B$$

- Assuming

$$z_i \sim \mathcal{N}(z_i | \mu = u, \sigma^2)$$

where u is the true difference in error function we have shown:

$$p(u|D) = p_{\mathcal{T}}(u | \nu = n - 1, \mu = \hat{z}, \sigma = \tilde{\sigma})$$

- Therefore, we can test the hypothesis

H_0 : Model $\mathcal{M}_A$ and $\mathcal{M}_B$ have the same performance, $u = 0$

H_1 : Model $\mathcal{M}_A$ and $\mathcal{M}_B$ have different performance, $u \neq 0$.

- A p -value can be computed as

$$p = 2\text{cdf}_{\mathcal{T}}(-|\hat{z}| \mid \nu = n - 1, \mu = 0, \sigma = \tilde{\sigma}).$$



Which type of cross-validation?

- When using **setup I** choose K as large as feasible (leave-one-out)
- Hold-out has the benefit the training/test data is fixed
- Results will be significant with enough data
 - Focus on estimated effect size
 - Multiple-comparison problem
 - **Transparency, availability of datasets/code, breadth of testing, self-criticism** guarantees *reproducibility* not a sophisticated test
- In **setup II**, correlation of training data is taken into account and K -fold is optimal
 - Your **setup I** results do not generalize beyond your training data

Class imbalance and AUC

Class imbalance problem

- Many data sets have **imbalanced class distributions**
 - Example: Detection of defects that only occur rarely (e.g. 1/1,000,000)
 - Danger: Algorithm that says nothing is defect will be 99.999% correct
- **Solution approaches**
 - Resample to balance data sets
 - Modify existing classification algorithms
 - Measure performance in a way that takes balance into account

Resampling balanced data

- New sample has equal number of data objects from each class
- Approaches:
 - Undersampling: majority class: Throws out potentially useful data
 - Oversampling minority class: Increase data size and computational burden
 - Somewhere in between...

Imbalanced training data	1	2	3	4	5	6	7	8	9	10
Oversampling	1	2	3	4	5	7	9	10	6	6
	6	6	8	8	8	8				
Undersampling	3	5	6	8						
Somewhere in between	3	5	4	3	9	6	6	8	8	8

Confusion matrix

		<i>Predicted</i>	
		<i>positive</i>	<i>negative</i>
<i>A c t u a l</i>	<i>positive</i>	TP True Positive	FN False Negative
	<i>negative</i>	FP False Positive	TN True Negative

Precision and recall

- **Precision**

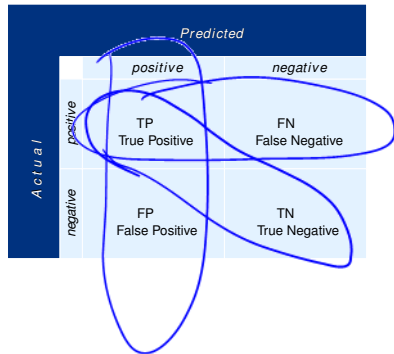
- Fraction of true positive among objects predicted to be positive

$$p = \frac{TP}{TP+FP}$$

- **Recall**

- Fraction of objects predicted to be positive among all positive objects

$$r = \frac{TP}{TP+FN}$$



Quiz 1: Precision/Recall

Consider two different classifiers, and suppose on a test set with 20 positive observations:

- Classifier 1 detects 54 positive of which 18 are actually positive
- Classifier 2 detects 16 positive of which 14 are actually positive

		Predicted		
		positive	negative	
Actual	positive	18 TP True Positive	14 FN False Negative	2
	negative	36 FP False Positive	2 TN True Negative	6
		54	16	20

• Precision

- Fraction of true positive among objects predicted to be positive

$$p = \frac{TP}{TP + FP}$$

• Recall

- Fraction of objects predicted to be positive among all positive objects

$$r = \frac{TP}{TP + FN}$$

What is the precision and recall of the two classifiers?

- A. Classifier 1: $p_1 = \frac{2}{3}, r_1 = \frac{7}{10}$
 Classifier 2: $p_2 = \frac{1}{3}, r_2 = \frac{1}{3}$
- B. Classifier 1: $p_1 = \frac{1}{3}, r_1 = \frac{9}{10}$
 Classifier 2: $p_2 = \frac{1}{3}, r_2 = \frac{9}{10}$
- C. Classifier 1: $p_1 = \frac{2}{3}, r_1 = \frac{7}{10}$
 Classifier 2: $p_2 = \frac{1}{3}, r_2 = \frac{9}{10}$
- D. Classifier 1: $p_1 = \frac{1}{3}, r_1 = \frac{9}{10}$
 Classifier 2: $p_2 = \frac{7}{8}, r_2 = \frac{7}{10}$

Which classifier would you use if the objective was to detect credit-card fraud (the positive class corresponds to fraud)

Solution:

The precision is the number of true positives divided by the number *predicted* to be positive. I.e. for the two classifiers

$$p_1 = \frac{18}{54} = \frac{1}{3}, \quad p_2 = \frac{14}{16} = \frac{7}{8}.$$

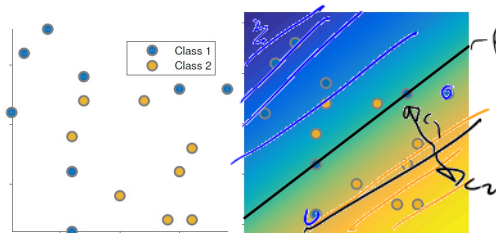
The recall is the number of true positives divided by the number of *actual* positives:

$$r_1 = \frac{18}{20} = \frac{9}{10}, \quad r_2 = \frac{14}{20} = \frac{7}{10}.$$

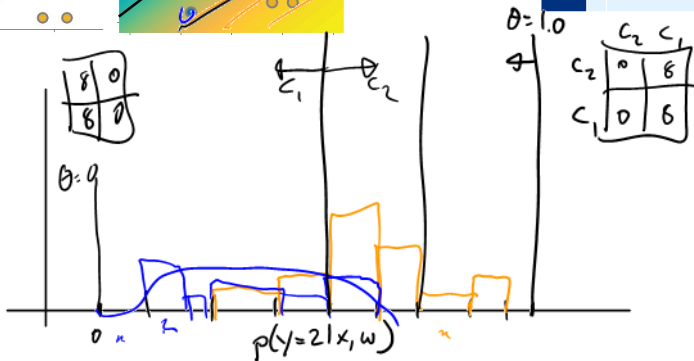
If the classifiers are supposed to detect credit-card fraud, we probably want a classifier which does not incorrectly *miss* many fraudulent transactions, i.e. it should have a low value of FN. In that case a high recall is desirable, and we should go with classifier 1.

		Predicted		
		positive	negative	
Actual	positive	18 TP True Positive	2 FN False Negative	20
	negative	36 FP False Positive	2 TN True Negative	20
		54	16	

Classification threshold



		Predicted	
		positive	negative
Actual	positive	TP True Positive	FN False Negative
	negative	FP False Positive	TN True Negative

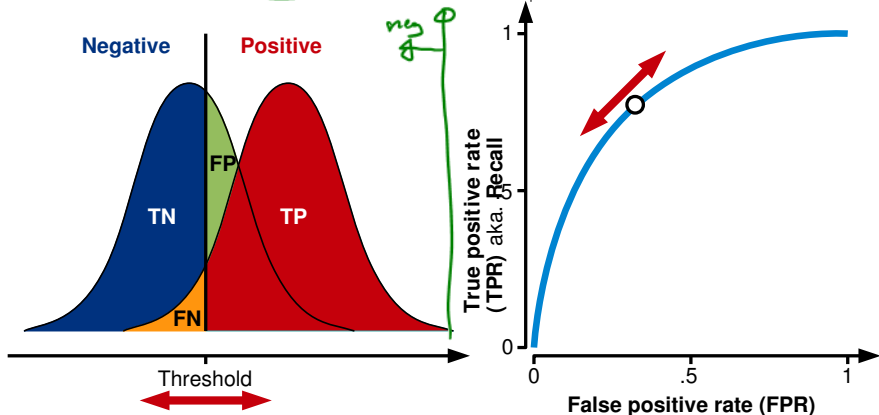


Receiver operating characteristic (ROC)

$$\begin{array}{c|c} 0 & N_0 \\ \hline 0 & N_1 \end{array}$$

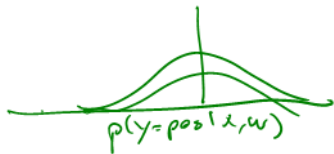
	pos	neg
pos	TP	FN
neg	FP	TN

$$TPR = \frac{TP}{TP + FN}$$



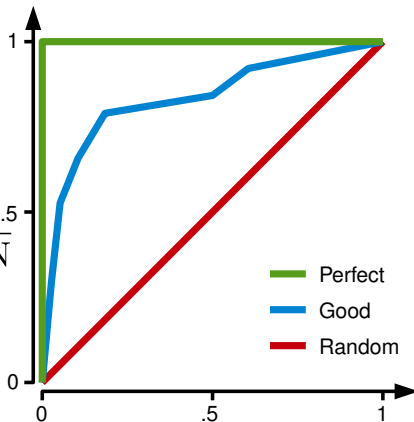
$$FPR = \frac{FP}{TN + FP}$$

Receiver operating characteristic (ROC)



True positive rate
aka. Recall

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$



False positive rate

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

	pos	neg
pos	TP	FN
neg	FP	TN

Quiz2: AUC

neg.
pos.

	3 gears ($x_5 = 3$)	4 gears ($x_5 = 4$)	5 gears ($x_5 = 5$)
Low mpg ($y = 0$)	13	2	2
High mpg ($y = 1$)	2	10	3

Table 1: Number of low mpg and high mpg cars (i.e. $y = 0$ and $y = 1$) according to the number of gears, i.e. $x_5 = 3$, $x_5 = 4$, or $x_5 = 5$.

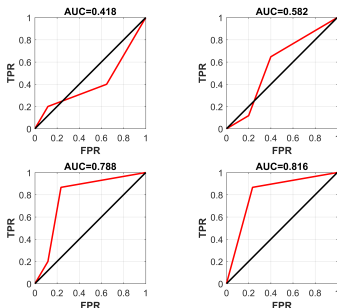


Figure 1: Four different receiver operator characteristic (ROC) curves and their area under curve (AUC) value.

A dataset representing cars contain an attribute x_5 corresponding to the number of gears. We wish to evaluate how well the number of gears predict low mpg, ($y = 0$, considered the negative class) from high mpg, ($y = 1$, considered the positive class) based on the data given in Table 1. For this purpose, we will evaluate the area under curve (AUC) of the receiver operator characteristic (ROC) using the feature x_5 . Which one of the ROC curves given in Figure 1 corresponds to using x_5 to discriminate between low mpg ($y = 0$) and high mpg ($y = 1$)?

- The curve having AUC=0.418
- The curve having AUC=0.582
- The curve having AUC=0.788
- The curve having AUC=0.816
- Don't know.

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

(Hint: Select a value e.g. $x_5 = 4$. We then predict cars with 4 or more gears as being in the positive class and otherwise negative. Compute the FPR and TPR using this prediction and use the (FPR, TPR) values to discriminate between the curves)

	pos	neg
pos	TP	FN
neg	FP	TN

Solution:

solution:

$$\theta = 2.5$$

	pos	neg
pos	15	0
neg	17	0

(FPR, TPR)

$$\left(\frac{17}{17}, \frac{15}{15}\right)$$

$$(1, 1)$$

$$\theta = 3.5$$

	pos	neg
pos	13	2
neg	4	13

$$\left(\frac{4}{17}, \frac{13}{15}\right)$$

$$(0.24, 0.87)$$

$$\theta = 4.5$$

	pos	neg
pos	3	12
neg	2	15

$$\left(\frac{2}{17}, \frac{3}{15}\right)$$

$$(0.12, 0.2)$$

$$\theta = 5.5$$

	pos	neg
pos	0	15
neg	0	17

$$\left(\frac{0}{17}, \frac{0}{15}\right)$$

$$(0, 0)$$

The ROC curve can be calculated by lowering the threshold, as no cars have more than 5 forward gears a threshold above 5 will result in the point (0,0). Lowering the threshold we find at the value 5 that 2/17 of the low mpg cars (FPR) are at 5 and 3/15 of the high mpg cars (TPR) are at 5 corresponding to

the point (2/17, 3/15). When lowering to a threshold of 4 gears or more we are at the point (4/17, 13/15) and at a threshold at 3 gears or more we have (1,1). Thus, this curve corresponds to the curve having $AUC=0.788$.

Summary

Resources

<https://www.youtube.com> Video tutorial on ROC curve and AUC

(<https://www.youtube.com/watch?v=0A16eAyP-yo>)

<https://machinelearningmastery.com> Statistical comparison of the cross-validation estimate of the generalization error is not a solved problem. This reference provides an overview of various issues and proposed solutions. Note no simple solution exists.

([https://](https://machinelearningmastery.com/statistical-significance-tests-for-comparing-machine-learning-algorithms/)

machinelearningmastery.com/statistical-significance-tests-for-comparing-machine-learning-algorithms/)